

Introduction to Linear Regression

Linear regression is a statistical method used to establish a relationship between a dependent variable (target) and one or more independent variables (predictors). The primary goal is to predict the value of the target variable based on the predictor variables.

Types of Linear Regression

1. **Simple Linear Regression:** Involves one independent variable.

- **Formula:**

$$y = mx + c$$

- **Where:**
 - y = predicted value (dependent variable)
 - m = slope of the regression line
 - x = independent variable
 - c = y-intercept

2. **Multiple Linear Regression:** Involves multiple independent variables.

- **Formula:**

$$y = m_1x_1 + m_2x_2 + \dots + m_nx_n + c$$

- **Where:**
 - y = predicted value
 - $m_1, m_2, \dots, m_n$ = coefficients for each predictor
 - $x_1, x_2, \dots, x_n$ = independent variables

R-squared (R^2)

- Represents the proportion of variance in the dependent variable that can be explained by the independent variables.
- **Formula:**

$$R^2 = 1 - \frac{RSS}{TSS}$$

- **Where:**
 - RSS = Residual Sum of Squares
 - TSS = Total Sum of Squares

Adjusted R-squared

- Adjusted for the number of predictors in the model.
- Formula:**

$$AdjustedR^2 = 1 - \left(\frac{(1 - R^2)(N - 1)}{N - p - 1} \right)$$

- Where:
 - N = number of observations
 - p = number of predictors

Residual Sum of Squares (RSS)

- Measures the level of variance in error.
- Formula:**

$$RSS = \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Mean Squared Error (MSE)

- Average of the squares of errors.
- Formula:**

$$MSE = \frac{RSS}{N}$$

Root Mean Squared Error (RMSE)

- Square root of MSE; provides error in the same units as the outcome variable.
- Formula:**

$$RMSE = \sqrt{MSE} = \sqrt{\frac{RSS}{N}}$$

Evaluation Metrics for Regression Tasks

When evaluating regression models, several metrics are commonly used:

- Mean Absolute Error (MAE):**

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

- Residual Standard Error (RSE):**

$$RSE = \sqrt{\frac{RSS}{N - 2}}$$

Evaluation Metrics for Classification Tasks

For classification tasks, evaluation metrics include:

- **Confusion Matrix:**

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall (Sensitivity)

$$Recall = \frac{TP}{TP + FN}$$

F1 Score

The harmonic mean of precision and recall.

$$F1Score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

Difference Between R-squared and Adjusted R-squared

- **R-squared (R^2)** measures how much variance in the dependent variable is explained by independent variables. It can increase with more predictors, even if those predictors are not meaningful.
- **Adjusted R-squared** adjusts for the number of predictors in the model. It only increases if adding a new predictor improves the model more than would be expected by chance.

1. What is Gradient Descent?

Gradient descent is an iterative first-order optimization algorithm that aims to find the local minimum of a differentiable function. It works by taking steps proportional to the negative of the gradient (or approximate gradient) of the function at the current point. This method is particularly useful for minimizing cost functions in various machine learning models, such as linear regression and neural networks.

Key Concepts:

- **Cost Function:** A function that measures how well a model's predictions match the actual data. The goal of gradient descent is to minimize this function.
- **Gradient:** The vector of partial derivatives of the cost function with respect to its parameters, indicating the direction of steepest ascent.
- **Learning Rate (η):** A hyperparameter that controls how much to change the model parameters with respect to the gradient. It determines the size of each step taken towards the minimum.

2. How Does Gradient Descent Work?

The basic idea behind gradient descent is to start with an initial guess for the parameters and iteratively update them based on the gradient of the cost function. The update rule can be expressed mathematically as follows:

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \cdot \nabla J(\theta)$$

Where:

- θ_{new} = updated parameter values
- θ_{old} = current parameter values
- η = learning rate
- $\nabla J(\theta)$ = gradient of the cost function with respect to θ

Steps in Gradient Descent:

1. **Initialize Parameters:** Start with random values for model parameters.
2. **Compute Gradient:** Calculate the gradient of the cost function at the current parameter values.
3. **Update Parameters:** Adjust parameters using the update rule.
4. **Repeat:** Continue steps 2 and 3 until convergence (when changes become negligible) or until a maximum number of iterations is reached.

3. Types of Gradient Descent

There are several variants of gradient descent, each suited for different scenarios:

3.1 Batch Gradient Descent

- Computes the gradient using the entire dataset.
- Pros: Converges smoothly; guarantees convergence to global minimum for convex functions.
- Cons: Can be slow and memory-intensive for large datasets.

3.2 Stochastic Gradient Descent (SGD)

- Computes the gradient using only one training example at a time.
- Pros: Faster updates; can escape local minima due to its noisy nature.
- Cons: More fluctuations in convergence; may not settle down at a minimum.

3.3 Mini-Batch Gradient Descent

- A compromise between batch and stochastic methods; computes gradients on small batches of data.
- Pros: Reduces variance in parameter updates; faster than batch GD while maintaining stability.

4. Learning Rate

The learning rate is crucial for effective training:

- **Too High:** The algorithm may overshoot and diverge, failing to converge.
- **Too Low:** The training process will be slow, potentially getting stuck before reaching convergence.

5. Cost Function

The cost function measures how well your model's predictions match actual outcomes. For example, in linear regression, it can be defined as:

$$J(w, b) = \frac{1}{n} \sum_{i=1}^n (y_i - (wx_i + b))^2$$

Where:

- y_i = actual output
- $w x_i + b$ = predicted output
- n = number of samples

6. Convergence Criteria

Gradient descent continues until one or more stopping criteria are met:

1. The maximum number of iterations is reached.
2. The change in cost function between iterations falls below a predefined threshold.

Assumptions of Linear Regression

1. **Linearity:** A linear relationship exists between dependent and independent variables.
2. **Normality:** The residuals should be normally distributed.
3. **Homoscedasticity:** The variance of residuals should remain constant across all levels of independent variables.
4. **No Multicollinearity:** Independent variables should not be highly correlated with each other.
 - Variance Inflation Factor (VIF):
 - If $VIF < 5$: No multicollinearity
 - If $VIF > 10$: Multicollinearity present
5. **Independence:** Observations should be independent.

Bias-Variance Tradeoff

- **Low Bias & High Variance:** Overfitting (model learns noise).
- **High Bias & Low Variance:** Underfitting (model is too simple).
- **Low Bias & Low Variance:** Generalized model.

Difference Between Scaling and Variance

- **Scaling** refers to transforming features to a similar scale to ensure that no single feature dominates others during model training. Common techniques include Min-Max scaling and Standardization.
- **Variance**, on the other hand, measures how much a set of numbers varies from their average value. In regression analysis, high variance may indicate overfitting.